

HW 1: Schrödinger Equation(s)

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Question 1) Compatible observables are those for which

$$[\hat{A}, \hat{B}] = 0, \quad (1)$$

and incompatible observables are those for which

$$[\hat{A}, \hat{B}] \neq 0. \quad (2)$$

You already know that position and momentum are incompatible observables, since

$$[\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}\mathbf{1} \text{ for the individual } i = x, y, z \text{ components.} \quad (3)$$

- Prove that a Hermitian operator has i) real eigenvalues *and* ii) eigenvectors associated with different eigenvalues are orthogonal.
- Prove that the eigenstates of $\hat{A}$ are also eigenstates of $\hat{B}$ if $\hat{A}$ and $\hat{B}$ are compatible observables. You can assume non-degenerate eigenvalues for $\hat{A}$. When the eigenbasis of $\hat{A}$ forms a complete basis, this proves that compatible observables can be simultaneously diagonalized.
- Show that if $\hat{A}$ and $\hat{B}$ are compatible observables and $\hat{A}$ and $\hat{C}$ are compatible observables, then $\hat{B}$ and $\hat{C}$ are also compatible observables, assuming non-degenerate eigenvalues for $\hat{A}$.
- As a special case of the above, show that if $\hat{A}$ and $\hat{B}$ are compatible observables, then any function of $\hat{A}$, say $f(\hat{A})$, is also compatible with $\hat{B}$. Assume non-degenerate eigenvalues for $\hat{A}$, $f(\hat{A})|a\rangle = f(a)|a\rangle$, and that $f(\hat{A})$ is Hermitian.

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Question 2) The time-dependent Schrödinger equation in operator, basis-independent form is

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle,$$

where $\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}})$. As we will show later, the momentum operator $\hat{\mathbf{p}}$ acts on position wavefunctions as $-i\hbar\nabla$. To be more precise, we have

$$\langle \mathbf{x} | \hat{\mathbf{p}} = -i\hbar\nabla \langle \mathbf{x} | \text{ and } \hat{\mathbf{p}} | \mathbf{x} \rangle = +i\hbar\nabla | \mathbf{x} \rangle.$$

A key manipulation is how to apply the momentum operator on a state $|\psi\rangle$ in the position representation:

$$\begin{aligned} \langle \mathbf{x} | \hat{\mathbf{p}} | \psi \rangle &= -i\hbar\nabla_{\mathbf{x}} \langle \mathbf{x} | \psi \rangle = -i\hbar\nabla_{\mathbf{x}} \psi(\mathbf{x}), \\ \langle \psi | \hat{\mathbf{p}} | \mathbf{x} \rangle &= +i\hbar\nabla_{\mathbf{x}} \langle \psi | \mathbf{x} \rangle = +i\hbar\nabla_{\mathbf{x}} \psi^*(\mathbf{x}). \end{aligned} \quad (4)$$

Note that these are complex conjugates of each other as it must be for a Hermitian operator. Pay close attention to what ∇ is differentiating with respect to and to the minus signs here.

- derive the time-dependent Schrödinger equation for the energy wavefunction $\psi_n(t) = \langle n | \psi(t) \rangle$ for a discrete spectrum, and then again for $\psi(E, t) = \langle E | \psi(t) \rangle$ in a continuous

spectrum. Note that $\hat{H}(t) |n\rangle = E_n |n\rangle$ for a discrete spectrum and $\hat{H} |E\rangle = E |E\rangle$ for a continuous spectrum.

By spectrum I mean the set of all possible energy eigenvalues of the Hamiltonian operator. Note that in this language, E_n and $|n\rangle$ are time-independent.

- b) Derive the time-dependent Schrödinger equation in position space (in 3 spatial dimensions), i.e., find the differential equation obeyed by the wavefunction $\psi(\mathbf{x}, t) = \langle \mathbf{x} | \psi(t) \rangle$.

Use a Hamiltonian of the form $\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}})$ and a potential that depends only on position, that is, $V(\hat{\mathbf{x}}) |\mathbf{x}\rangle = V(\mathbf{x}) |\mathbf{x}\rangle$ with $V(\mathbf{x})$ being just a number for each position vector $\mathbf{x}$.



Bonus) Repeat the same steps to derive the time-dependent Schrödinger equation in momentum space, i.e., find the *integro-differential* equation obeyed by $\phi(\mathbf{p}, t) = \langle \mathbf{p} | \psi(t) \rangle$.

The momentum and position kets are related by the Fourier transform, $\langle \mathbf{x} | \mathbf{p} \rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}$.

Your answer should be an integro-differential equation that depends on the Fourier transform of the potential:

$$\langle \mathbf{p} | \hat{V} | \mathbf{p}' \rangle = \tilde{V}(\mathbf{p} - \mathbf{p}') = \frac{1}{(2\pi\hbar)^3} \int d^3\mathbf{x} e^{-i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar} V(\mathbf{x}).$$

Some useful relations:

$$\int d^3\mathbf{x} |\mathbf{x}\rangle \langle \mathbf{x}| = \hat{I}, \quad \int d^3\mathbf{p} |\mathbf{p}\rangle \langle \mathbf{p}| = \hat{I}, \quad (5)$$

$$\langle \mathbf{x} | \mathbf{p} \rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \text{ and its hermitian conjugate version} \quad (6)$$

$$(7)$$

$$\int d^3\mathbf{x} \langle \mathbf{p} | \mathbf{x} \rangle \langle \mathbf{x} | \mathbf{p}' \rangle = \int d^3\mathbf{x} \frac{1}{(2\pi\hbar)^3} e^{-i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar} = \delta(\mathbf{p} - \mathbf{p}') \quad (8)$$

$$\int d^3\mathbf{p} \langle \mathbf{x} | \mathbf{p} \rangle \langle \mathbf{p} | \mathbf{x}' \rangle = \int d^3\mathbf{p} \frac{1}{(2\pi\hbar)^3} e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{x}')/\hbar} = \delta(\mathbf{x} - \mathbf{x}') \quad (9)$$